

Seamless-walk: Novel Natural Virtual Reality Locomotion Method with a High-Resolution Tactile Sensor

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ABSTRACT

Natural movement is a challenging problem in virtual reality locomotion. However, existing foot-based locomotion methods lack naturalness due to physical limitations caused by wearing equipment. Therefore, in this study, we propose Seamless-walk, a novel virtual reality (VR) locomotion technique to enable locomotion in the virtual environment by walking on a high-resolution tactile carpet. The proposed Seamless-walk moves the user's virtual character by extracting the users' walking speed and orientation from raw tactile signals using machine learning techniques. We demonstrate that the proposed Seamless-walk is more natural and effective than existing VR locomotion methods by comparing them in VR game-playing tasks.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Virtual reality (VR) allows us to experience new interesting spaces by creating a virtual environment (VE). However, existing space-efficient locomotion methods, such as point & teleport and arm-swing, are different from the actual gait motion. They require users to use their hands to locomote, fostering unnatural conditions. Thus, there is a gap between real-world and VE movements, which can provide an unnatural feeling of movement [1].

To address this issue, several foot-based locomotion techniques have been proposed, such as thread mill [2,3] and shoe-type interface [6]. However, these methods force users to wear equipment; thereby, constraining the human body and causing inconvenience for the users [4]. For example, most thread mill systems use a vest or railing to constrain users. Additionally, users have to wear a foot-equipment to use the shoe-type interfaces.

In this study, we propose Seamless-walk, a novel VR locomotion technique, to enable the real world's gait action as a virtual world's locomotion input using high-resolution tactile carpets [5]. The proposed method enables users to locomote on VE through walking-in-place (WIP). It identifies the user's intentions by extracting the direction and speed of in-place walking from raw foot pressure image. We employed machine learning techniques to process raw sensory signals produced by tactile carpet. Unlike existing WIP-based VR locomotion methods, the proposed method allows users to freely adjust locomotion without constraining their bodies. Furthermore, we evaluated the proposed Seamless-walk method on VR game-playing tasks and compared the result with existing

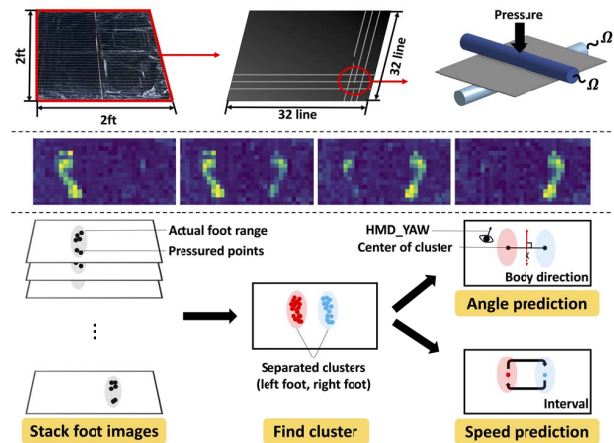


Figure 1: **Top:** Each sensing carpet tile is 2 ft x 2 ft in size with 32 x 32 sensors. Each sensor is located at the intersection. **Middle:** Sensing responses from sensing carpet when a person is walking. **Bottom:** The clustering technique specifies the left and right feet using sensing response images to extract user movement.

locomotion methods. The result shows that the proposed method performs better than other widely used locomotion techniques in terms of natural locomotion. It is also effective when using hands and feet simultaneously to play VR games.

2 METHOD

As illustrated in Figure 1 (top), our tactile sensing carpet is constructed by orthogonally aligning the electrodes on each side of the piezoresistive films. The sensing carpet enables measurement in real-time during the VR game, as shown in Figure 1 (middle). We use the proposed sensor and clustering technique to find the location of the user's feet (Foot1, Foot2) through pressured points in 14 consecutive sensing responses. Then, we find the body direction by taking the arctangent from the perpendicular slope of the straight line connecting the center point of each cluster (foot), as expressed in equation 1.

$$BD = \arctan\left(-\frac{Foot1_x - Foot2_x}{Foot1_y - Foot2_y}\right) \quad (1)$$

After that, choose the moving direction by referring to the HMD yaw direction, as shown in Figure 1 (bottom). Additionally, we calculate the movement speed using the user's foot switching interval time. If the current interval time (I_c) is longer than the upper bound interval time (I_u), set speed 0. Otherwise, adjust speed in proportion

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Table 1: Descriptive Statistics for NQ and RTLX.

	Arm-swing	Teleport	Seamless-walk
NQ			
Naturalness	5.20(2.34)	3.73(2.52)	6.73(2.34)
RTLX			
Effort	50.0(18.1)	44.0(22.0)	67.3(15.8)
Physical	42.0(20.8)	37.3(26.0)	61.3(16.4)

to the interval time difference ($I_u - I_c$) by multiplying standard speed ($speed_{standard}$), as expressed in equation 2.

$$speed = \begin{cases} 0, & \text{if } I_c > I_u \\ (I_u - I_c) \times speed_{standard}, & \text{Otherwise} \end{cases} \quad (2)$$

Calculated moving directions (angle) and speeds are used to manipulate virtual characters' locomotion.

3 EXPERIMENTS

Study design In the experiment, we compared three locomotion methods: teleport, arm-swing, and Seamless-walk. We chose teleport and arm-swing because they are the most commonly used VR locomotion methods. Additionally, they allow to control the direction without respect to viewpoint compared to other locomotion methods that determine the direction using viewpoint. The experiment order has been determined using Balanced Latin Square to reduce the learning effects. We conducted the experiment with 15 Korean college students (Mean age = 24.5, SD = 4.92, F = 5).

Game design The VR game was implemented using Unreal Engine 4. We employed first person shooter genre. The player should reach the goal safely while shooting the monsters with a gun. The environment of the VR game is night; thus, the players have to constantly use the flashlight to light surroundings with another hand. The distance to the goal is 300 m. Furthermore, the monsters are generated at five random locations among the 20 monster creation locations designated around the road. player should shoot monsters as soon as possible to avoid score loss.

Procedure First, the participants were informed about the purpose of the research, procedure of the experiment, rules of the game, and what kind of information was collected. Then, they were asked to fill out the pre-task questionnaire on demographics. After the questionnaire, they had 5 minutes to get used to the VR equipment and each locomotion method.

During the experiment, we recorded the number of missed enemies, game scores, and finish times. The total experimental time was about 40 minutes for a person. Whenever the participants complete the experiment using each locomotion method, the participants were asked to fill out the simulator sickness questionnaire (SSQ), raw NASA task load index (NASA-TLX) questionnaire, game experience questionnaire (GEQ), and the naturalness questionnaire (NQ) that we prepared to verify whether the Seamless-walk is more natural than other methods.

4 RESULTS

We conducted the ANOVA test on NQ, NASA-TLX, GEQ, SSQ questionnaires, and game statistics result to find group difference. Then, we conducted post-hoc tests, such as the HSD test for questionnaire data, Kruskal–Wallis test for in-game locomotion suitability rank data. Suitability rank has been decided for each participant through game completeness, score, and time lap order. As presented in Table 1, we obtain that the proposed method provides more naturalness than teleport from NQ (Locomotion method of the game was similar to real world) at a post-hoc test ($p = 0.004$). Additionally, in suitability aspect using the Kruskal–Wallis test, there were significant differences found between Seamless-walk and teleport,

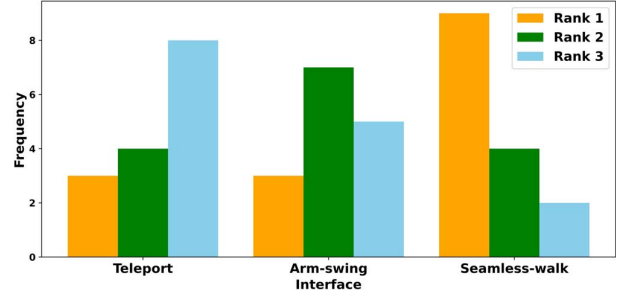


Figure 2: **Results of the in-game rank frequency.** The vertical axis is the rank frequency, and the horizontal axis is the interface. The first place is an orange bar; the second place is a green bar; the third place is a light blue bar.

arm-swing ($p = 0.027$, $p = 0.015$). However, the result of effort measurement in NASA-TLX shows that our method requires more effort than others at post-hoc analysis ($p = 0.046$ with arm-swing, $p = 0.01$ with teleport). Similarly, the physical measurement shows a significant difference in the post-hoc test ($p = 0.04$ with arm-swing, $p = 0.004$ with teleport). The ANOVA results show that there was no significant difference between Seamless-walk and other methods in GEQ and SSQ.

5 DISCUSSION AND CONCLUSION

The result shows that the proposed method outperforms teleport; it is also similar to the arm-swing method in terms of naturalness as in Table 1. Additionally, the proposed method is more suitable for hand-busy games. These results demonstrate that our method can provide users with more natural and effective locomotion experience than widely used locomotion methods in hand-busy games without reducing GEQ or SSQ scores. Furthermore, the proposed method is appropriate for people who want an active experience in VR games.

It is worth noting that the proposed method can be more improved by additional study. Additionally, deep learning could improve recognition of human intentions in future work since our tactile sensor provides high-dimensional sensory data.

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